

Sleep, Health and Life Style Report

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Contents

Introduction	2
Executive Summary.....	2
Project Objective	2
Key Insights.....	2
Immediate Future Actions.....	2
About the Dataset.....	3
Data Wrangling.....	3
Data Transformation	3
Exploratory Data Analysis	4
Exploratory Data Analysis Results	5
Data Visualizations and Insights.....	6
Charts	6
Dashboard	9
Key Data Insights	10
Considerations to keep in mind	10
Recommendations	11

Introduction:

Sleep, health, and lifestyle are deeply interconnected aspects of human well-being. Sleep is a vital biological function that allows the body and mind to rest, recover, and regenerate. It affects everything from brain function and emotional balance to immune strength and metabolism. Health, encompassing both physical and mental states, is influenced by various factors including diet, physical activity, stress levels, and, importantly, sleep quality and duration. Lifestyle refers to the daily habits and routines—such as work schedules, exercise, nutrition, and screen time—that shape how we live and, in turn, impact both sleep and health. Understanding the balance among these elements is essential for maintaining long-term well-being and preventing chronic illnesses.

The relationship between sleep, health, and lifestyle is cyclical and mutually reinforcing. A healthy lifestyle—marked by balanced nutrition, regular exercise, and stress management—can significantly improve sleep quality and duration. In turn, adequate and restful sleep enhances the body's ability to repair itself, boosts immunity, sharpens cognitive performance, and regulates mood, all of which contribute to better overall health. Conversely, poor lifestyle choices such as irregular routines, excessive screen time, substance use, or lack of physical activity can disrupt sleep patterns and lead to health issues like fatigue, anxiety, obesity, and heart disease. This dynamic interaction highlights the importance of maintaining a well-rounded lifestyle to support both sleep and health outcomes.

Executive Summary:

Project Objective:

The objective of this project is to perform a comprehensive analysis of sleep data, segmented by gender, occupation, and body mass index (BMI), while taking into account key health-related factors such as stress levels, physical activity, and heart rate.

Key Insights:

- Sleep duration and quality are closely linked, with longer sleep associated with better sleep quality and lower stress levels. Conversely, higher stress levels and shorter sleep durations negatively impact overall sleep health.
- Occupation, gender, age, and BMI significantly influence sleep patterns and disorders. For example, engineers sleep the most, scientists the least, females sleep longer than males, and sleep disorders are more common among overweight and obese individuals.
- Physical activity and daily steps have limited direct impact on sleep quality, though higher physical activity levels may slightly reduce the prevalence of sleep disorders, especially when combined with other healthy lifestyle factors.

Immediate Future Actions:

- Promote personalized, occupation- and demographic-specific sleep strategies by implementing workplace wellness programs, stress-reduction initiatives, and scheduling policies tailored to high-risk groups such as healthcare workers, scientists, males, and shift-based professionals.
- Support holistic sleep health by addressing contributing factors such as stress, poor sleep hygiene, and age-specific needs, emphasizing quality of sleep alongside duration through interventions like mindfulness, counseling, and environmental improvements.
- Integrate weight management and physical activity into broader sleep health efforts, recognizing that while BMI has a strong link to sleep disorders, physical activity alone has limited impact on sleep quality and should be complemented by preventive care and lifestyle adjustments. 244,188,156

About the Dataset:

The open-source dataset is taken from **Kaggle**.

This dataset contains sleep records of individuals, categorized by age, gender, body mass index (BMI), and occupation. It also incorporates key health and lifestyle factors such as sleep quality, physical activity levels, stress levels, blood pressure, and heart rate, providing a comprehensive view of the variables influencing sleep patterns.

Data Wrangling:

The csv data file was extracted using **readr package** in **R**. The Data Wrangling operations were performed using **tidyverse** in **R**. Some of the performed operations were:

- Missing Values verification
- Duplicate Verification
- Filtering Columns
- Correcting Data Types

Data Transformation:

Data Transformation was done by using **tidyverse** in **R**. Some of the performed operations were:

- Removing rows
- Dealing with Outliers
- Converting columns to factors
- Creation of specific views for visualizations

Exploratory Data Analysis:

Exploratory Data Analysis was done by using **tidyverse**, **ggplot2** and **corrplot** in R. Some of the performed operations were:

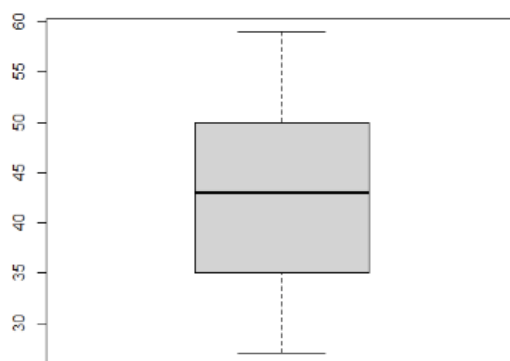
- Summary Statistics
- Arranging, Sorting and Grouping by different Statistics to better explore data
- Box plot Creation

Person.ID	Gender	Age	Occupation	Sleep.Duration
Min. : 1.00	Female:185	Min. :27.00	Nurse :73	Min. :5.800
1st Qu.: 94.25	Male :189	1st Qu.:35.25	Doctor :71	1st Qu.:6.400
Median :187.50		Median :43.00	Engineer :63	Median :7.200
Mean :187.50		Mean :42.18	Lawyer :47	Mean :7.132
3rd Qu.:280.75		3rd Qu.:50.00	Teacher :40	3rd Qu.:7.800
Max. :374.00		Max. :59.00	Accountant:37	Max. :8.500
			(other) :43	

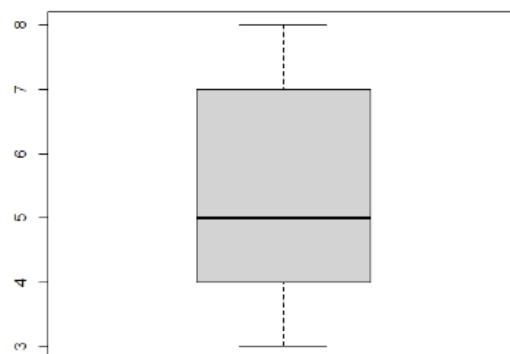
Quality.of.Sleep	Physical.Activity.Level	Stress.Level	BMI.Category
Min. :4.000	Min. :30.00	Min. :3.000	Length:374
1st Qu.:6.000	1st Qu.:45.00	1st Qu.:4.000	Class :character
Median :7.000	Median :60.00	Median :5.000	Mode :character
Mean :7.313	Mean :59.17	Mean :5.385	
3rd Qu.:8.000	3rd Qu.:75.00	3rd Qu.:7.000	
Max. :9.000	Max. :90.00	Max. :8.000	

Blood.Pressure	Heart.Rate	Daily.Steps	Sleep.Disorder	BMI.category
Length:374	Min. :65.00	Min. :3000	Length:374	Normal :195
Class :character	1st Qu.:68.00	1st Qu.:5600	Class :character	Normal weight: 21
Mode :character	Median :70.00	Median :7000	Mode :character	obese :10
	Mean :70.17	Mean :6817		Overweight :148
	3rd Qu.:72.00	3rd Qu.:8000		
	Max. :86.00	Max. :10000		

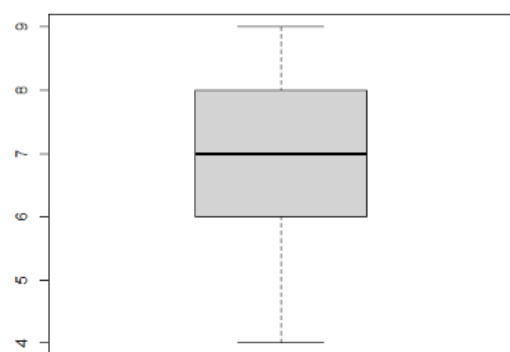
Sleep.disorder
Insomnia : 77
None :219
Sleep Apnea: 78



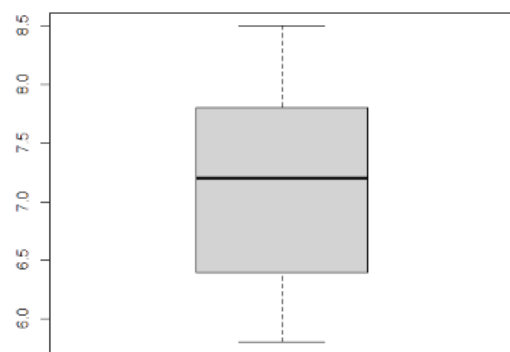
Age
Normally Distributed



Stress Level
Slightly Left Skewed



Quality of Sleep
Normally Distributed



Sleep Duration
Slightly Right Skewed

Exploratory Data Analysis Results:

1. The age distribution in the dataset demonstrates a normal pattern, indicating that the sample includes a balanced representation of individuals across various life stages. This symmetry around the central tendency suggests that there is no significant overrepresentation of either younger or older age groups, making the dataset suitable for drawing generalizable conclusions related to age-dependent sleep and health behaviors.
2. The distribution of stress levels is slightly left-skewed, meaning that a larger portion of individuals in the dataset report experiencing moderate to high levels of stress. This trend suggests that stress is a common factor affecting the population, and it may have a significant influence on other health indicators such as sleep quality and heart rate. The skew also highlights the need to consider stress management interventions in health-related recommendations.
3. The quality of sleep appears to follow a normal distribution, indicating that most individuals in the dataset experience average to good sleep quality. This balanced spread suggests that extreme values—either very poor or exceptionally high sleep quality—are less frequent. Such a distribution allows for reliable statistical analysis and reflects a generally healthy sleep profile within the population studied.
4. The distribution of sleep duration is slightly right-skewed, revealing that while most individuals get a moderate amount of sleep, there is a noticeable subset of the population that tends to sleep for longer periods. This rightward skew may be influenced by lifestyle factors, occupation types, or health conditions that necessitate more rest. Understanding this variation is important for assessing the impact of sleep duration on overall well-being.

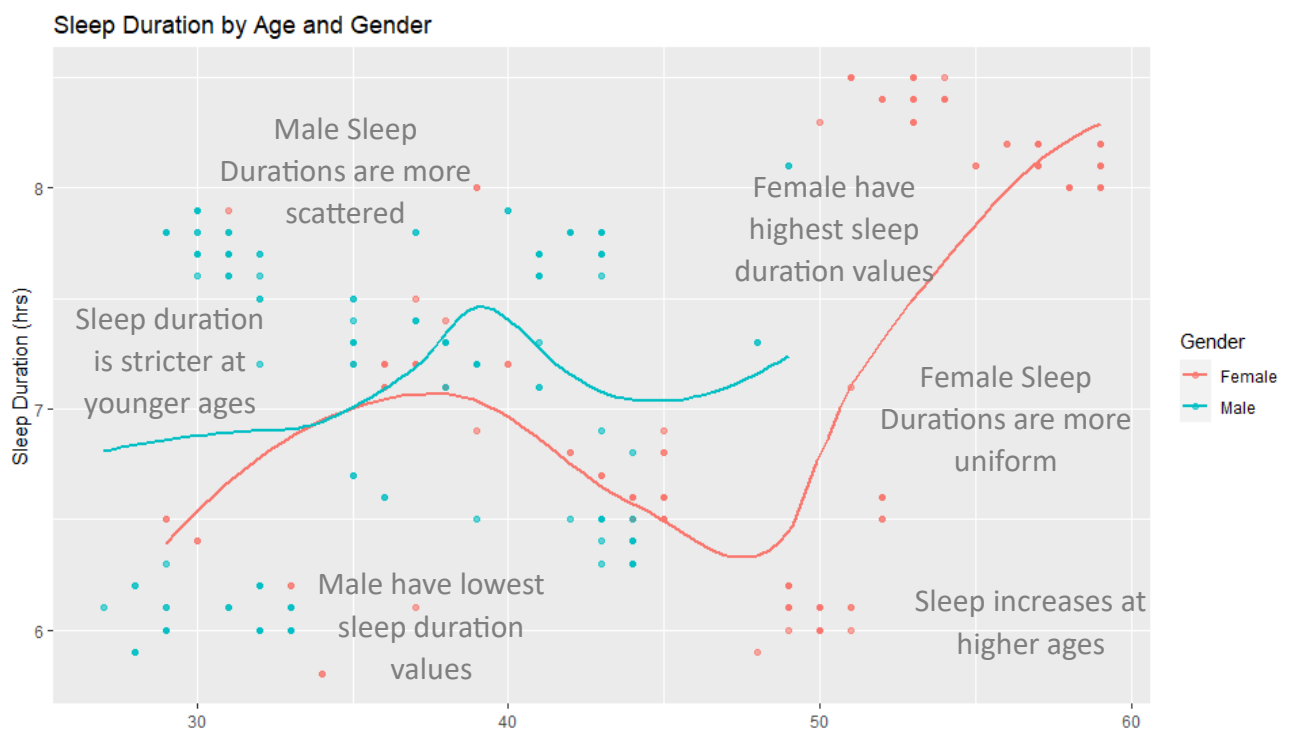
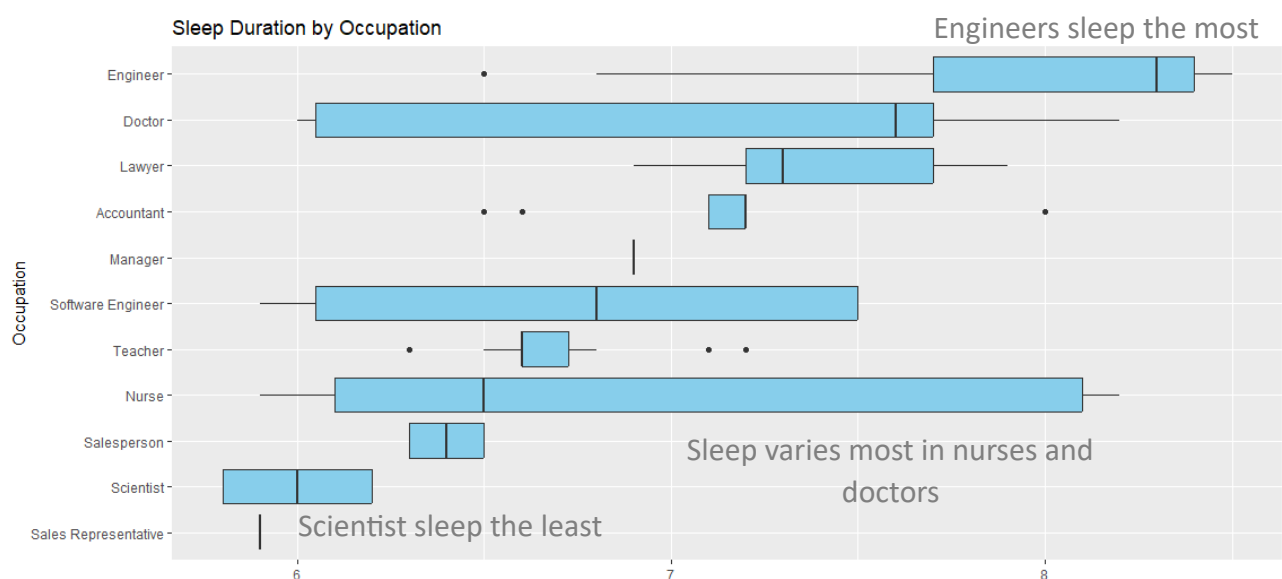
Data Visualizations and Insights:

5 specific visualizations were produced using **ggplot2**, **corrplot** and **plotly** in **R**. For a more thorough analysis, three dashboards were created using **ggplot2**, **plotly**, **flexdashboard** and the **Shiny Web Application** in **R**, featuring various visualizations. These interactive dashboards were designed to allow users to personalize their experience. Each visualization was made with a specific objective in mind. The most useful visualizations and dashboards containing the obtained insights are detailed below.

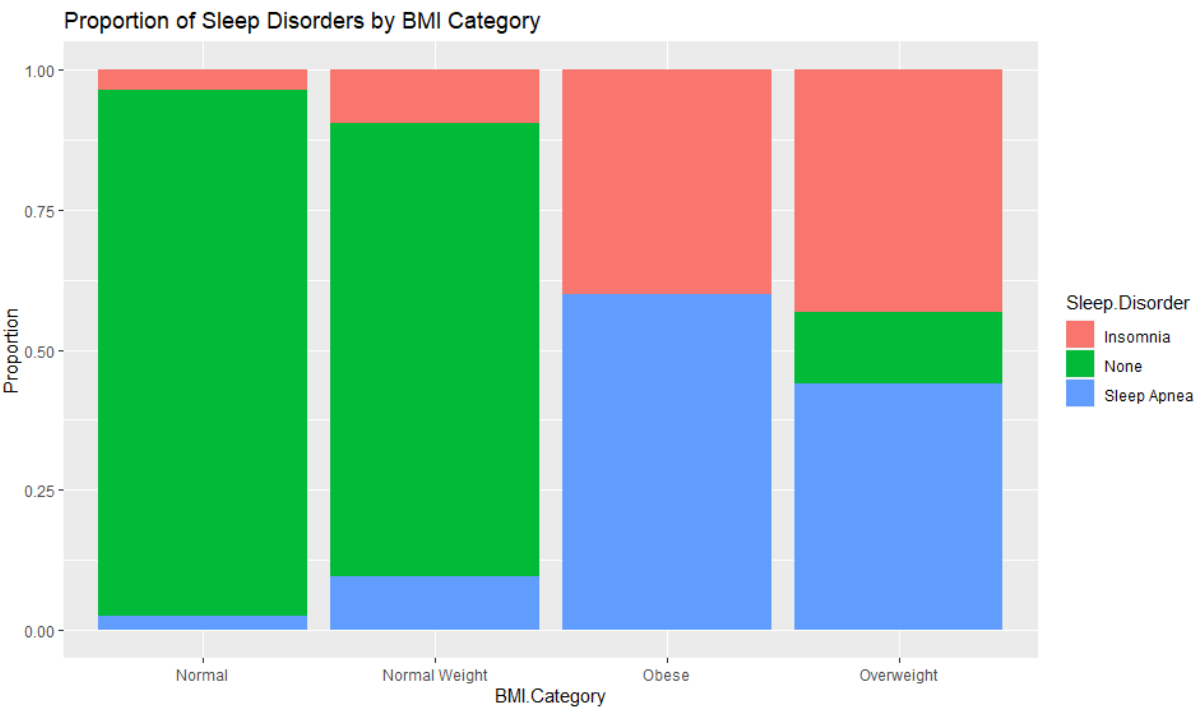
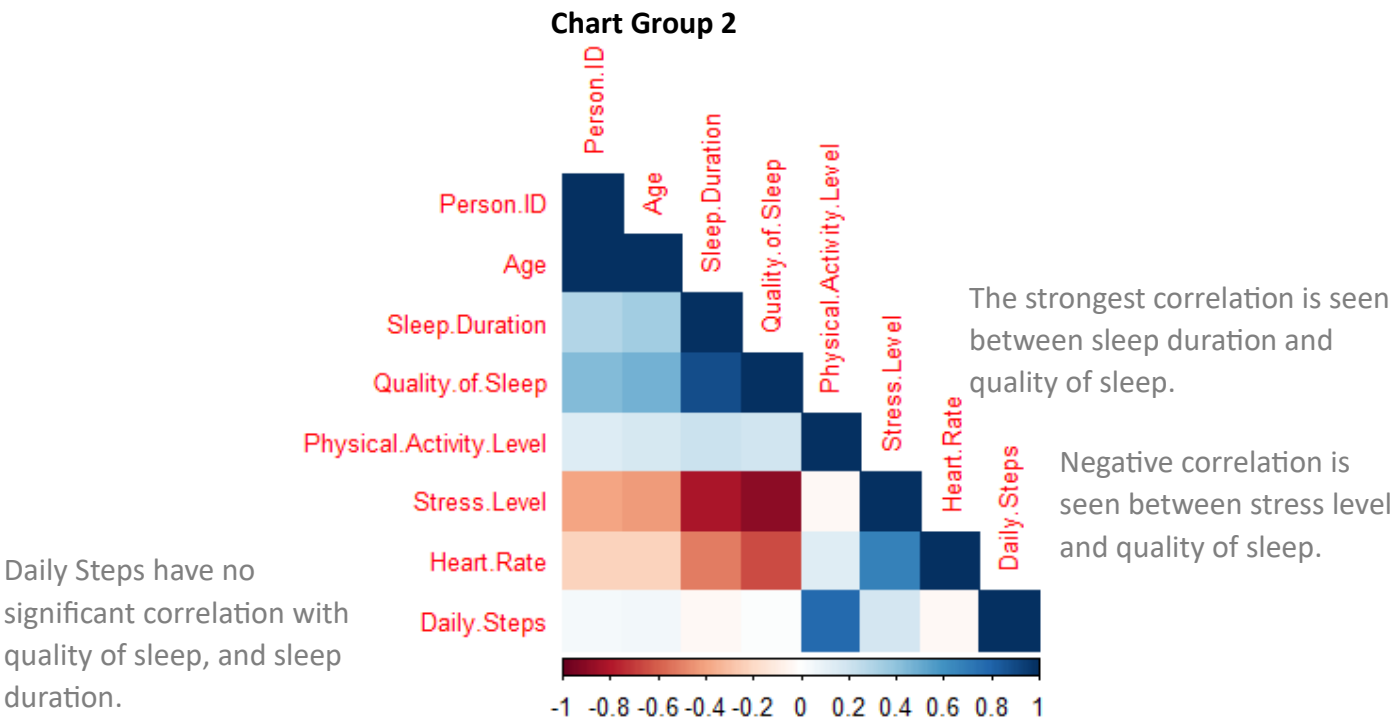
Charts:

1. The first visualization is showing the sleep duration by occupation, age, and gender.

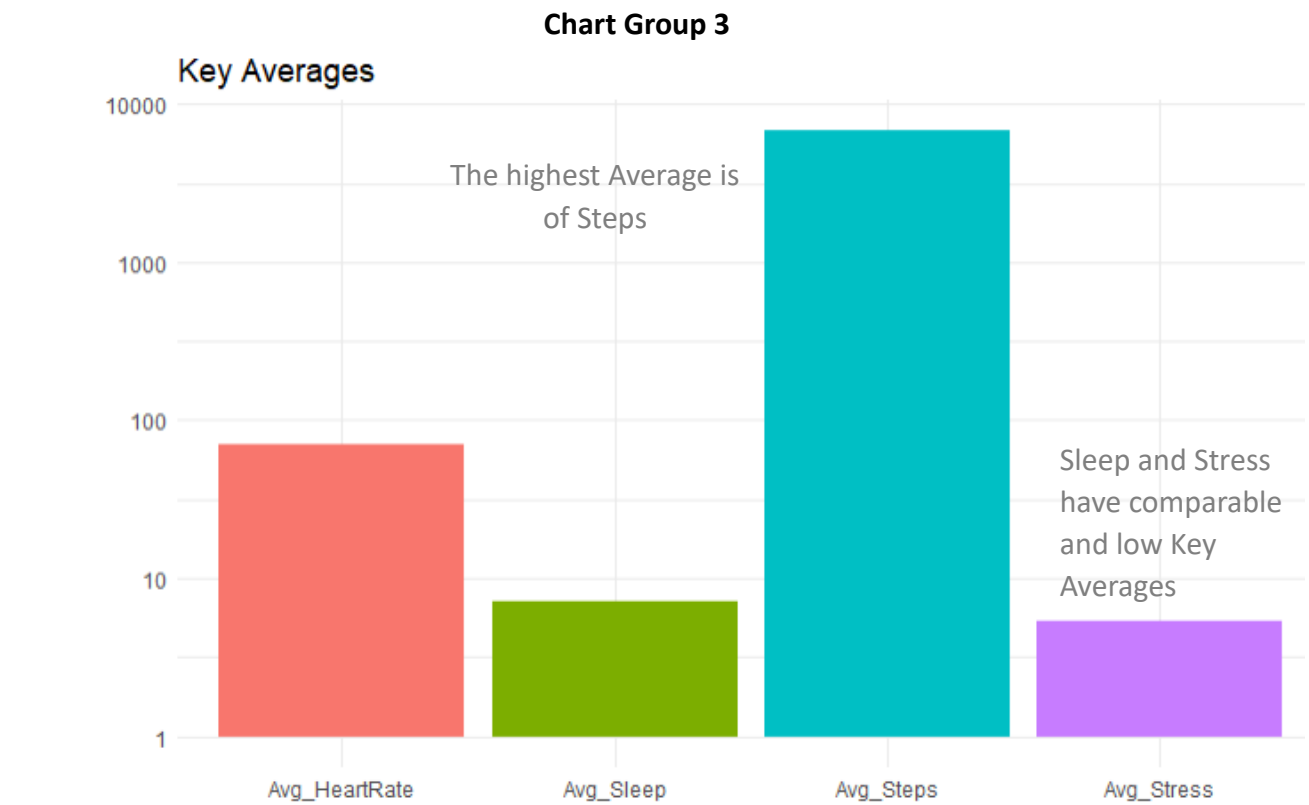
Chart Group 1



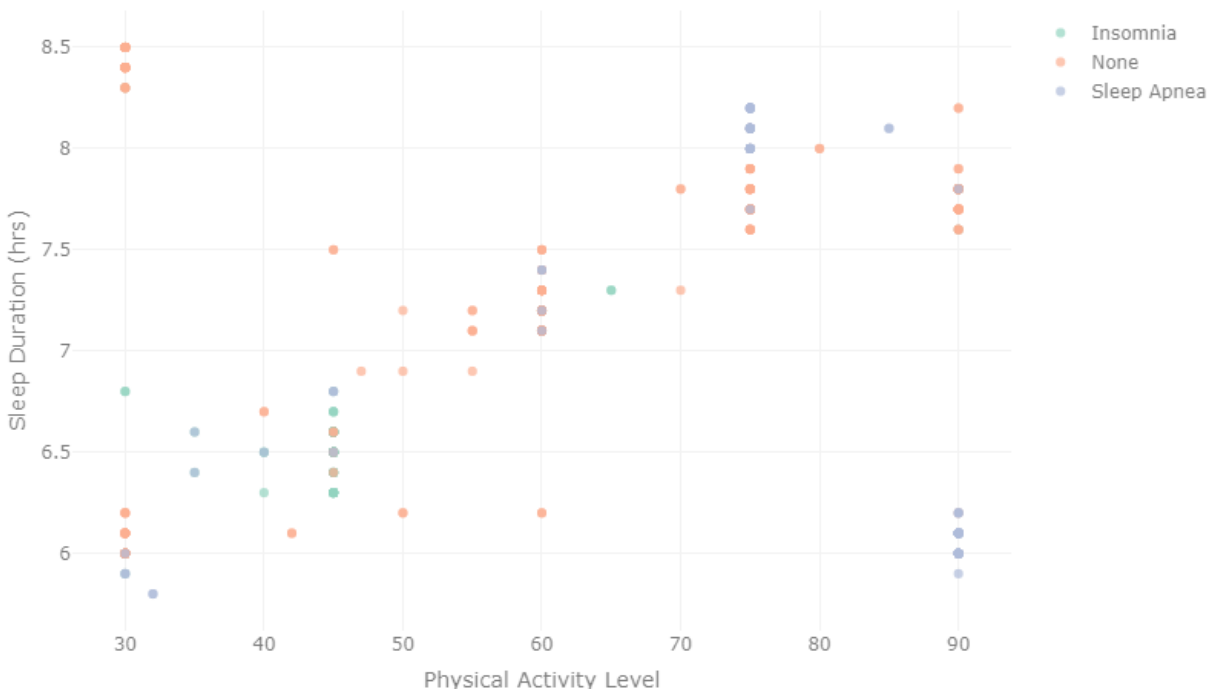
2. The second visualization is showing the correlation matrix between numerical quantities and proportion of sleep disorders by BMI category.



3. The second visualization is showing the correlation matrix between numerical quantities and proportion of sleep disorders by BMI category.



Physical Activity vs Sleep Duration by Sleep Disorder



People with high sleep durations have less sleep disorders

Increase in Physical Activity has some effect in reduction of sleep disorders

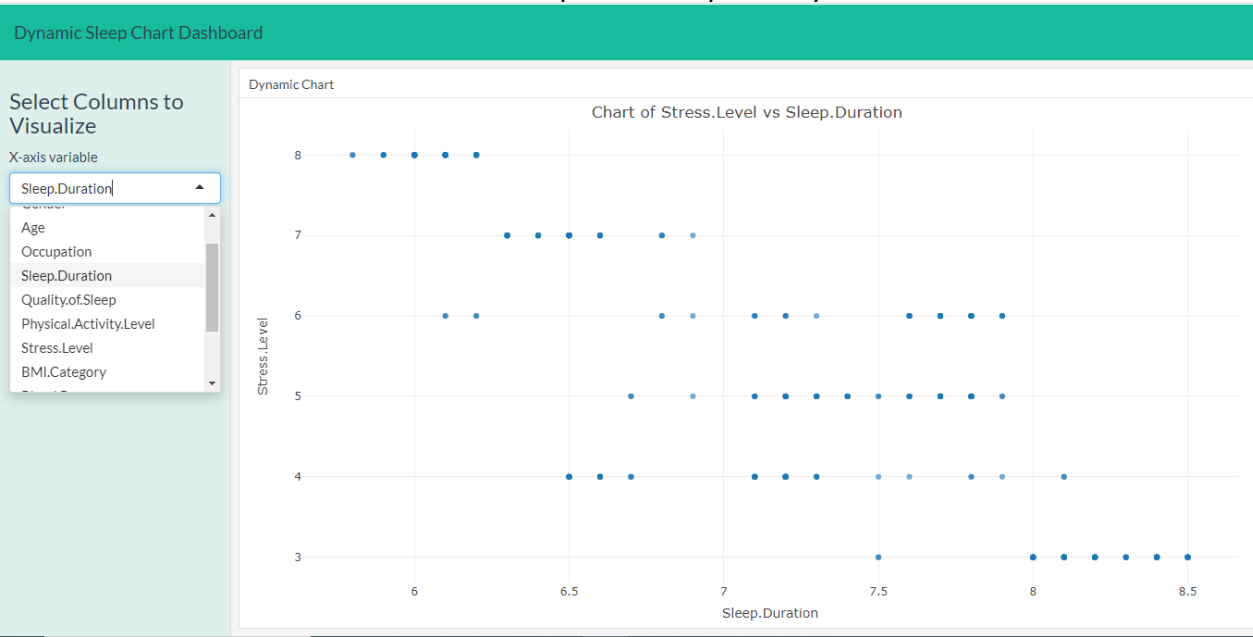
Dashboard:

The Dashboard visualizes categorical x-axis variables against numerical y-axis variables.

Dashboard



Higher Sleep Duration
improves Sleep Quality



Higher Sleep Duration
lowers stress level

Key Data Insights:

1. Engineering professionals exhibit the highest average sleep duration among all occupations.
2. The greatest variability in sleep duration is observed among healthcare professionals, particularly nurses and doctors.
3. Scientists consistently report the lowest average sleep duration compared to other occupational groups.
4. Male individuals display a wider range of sleep durations, indicating greater variability than their female counterparts.
5. Sleep duration appears to be more regulated and consistent in younger age groups, whereas older individuals tend to sleep for longer durations.
6. Females, on average, report the highest sleep durations, while males tend to have the lowest.
7. Sleep duration demonstrates the strongest positive correlation with quality of sleep, suggesting that longer sleep is associated with better sleep quality.
8. A notable negative correlation exists between stress levels and sleep quality, indicating that increased stress is associated with poorer sleep quality.
9. Daily step count shows no statistically significant relationship with either sleep duration or sleep quality.
10. Sleep disorders are predominantly found in individuals classified as overweight or obese based on BMI, highlighting a strong association with excess body weight.
11. Cases of insomnia and sleep apnea are relatively rare among individuals within the normal weight range.
12. Both stress level and sleep duration exhibit similarly low average values across the dataset.
13. An increase in physical activity level appears to contribute modestly to a reduction in the prevalence of sleep disorders.
14. Individuals with higher sleep durations tend to report fewer occurrences of sleep disorders.
15. Elevated sleep durations are associated with lower reported stress levels, suggesting a potential buffering effect of sufficient sleep against stress.

Considerations to keep in mind:

1. The dataset is not inclusive of all the details regarding sleep and lifestyle choices. As a result, the accuracy of this information is confined to the scope of the provided dataset.
2. Apart from the discussed metrics there are many other metrics that effect sleep and health.

Recommendations:

1. Encourage workplace wellness programs, especially in healthcare and scientific professions, where irregular schedules and mental fatigue contribute to shorter or highly variable sleep durations.
2. Since males exhibit greater variability and lower average sleep durations, targeted interventions (e.g., stress reduction, bedtime consistency) could be developed specifically for male populations.
3. Given that younger individuals tend to have stricter and more consistent sleep patterns, while older adults sleep longer, sleep guidelines should reflect age-related physiological needs rather than a uniform recommendation.
4. With the strongest positive correlation observed between sleep duration and quality, initiatives should emphasize improving the conditions that support quality sleep (e.g., noise reduction, sleep hygiene).
5. Since higher stress levels degrade sleep quality, integrate mindfulness, counseling, or workload adjustments into environments with chronic stressors, such as hospitals and corporate settings.
6. While beneficial for general health, daily steps and activity levels show minimal direct impact on sleep duration or quality. Thus, physical activity should be part of a broader sleep strategy rather than the sole focus.
7. With sleep disorders highly prevalent in overweight and obese individuals, promoting healthy weight through diet and exercise can significantly reduce the burden of insomnia and sleep apnea.
8. Continue to support and monitor individuals within the normal BMI range, as they exhibit the lowest incidence of sleep disorders. Preventive care can maintain this status.
9. Individuals with longer sleep durations report lower stress, suggesting that sleep should be prioritized in mental health and burnout prevention strategies.
10. Develop occupation-specific rest and scheduling policies, especially for shift-based roles (e.g., nurses, engineers), to promote healthy sleep cycles and reduce health risks.